1 | INTRODUCTION

Na Ma

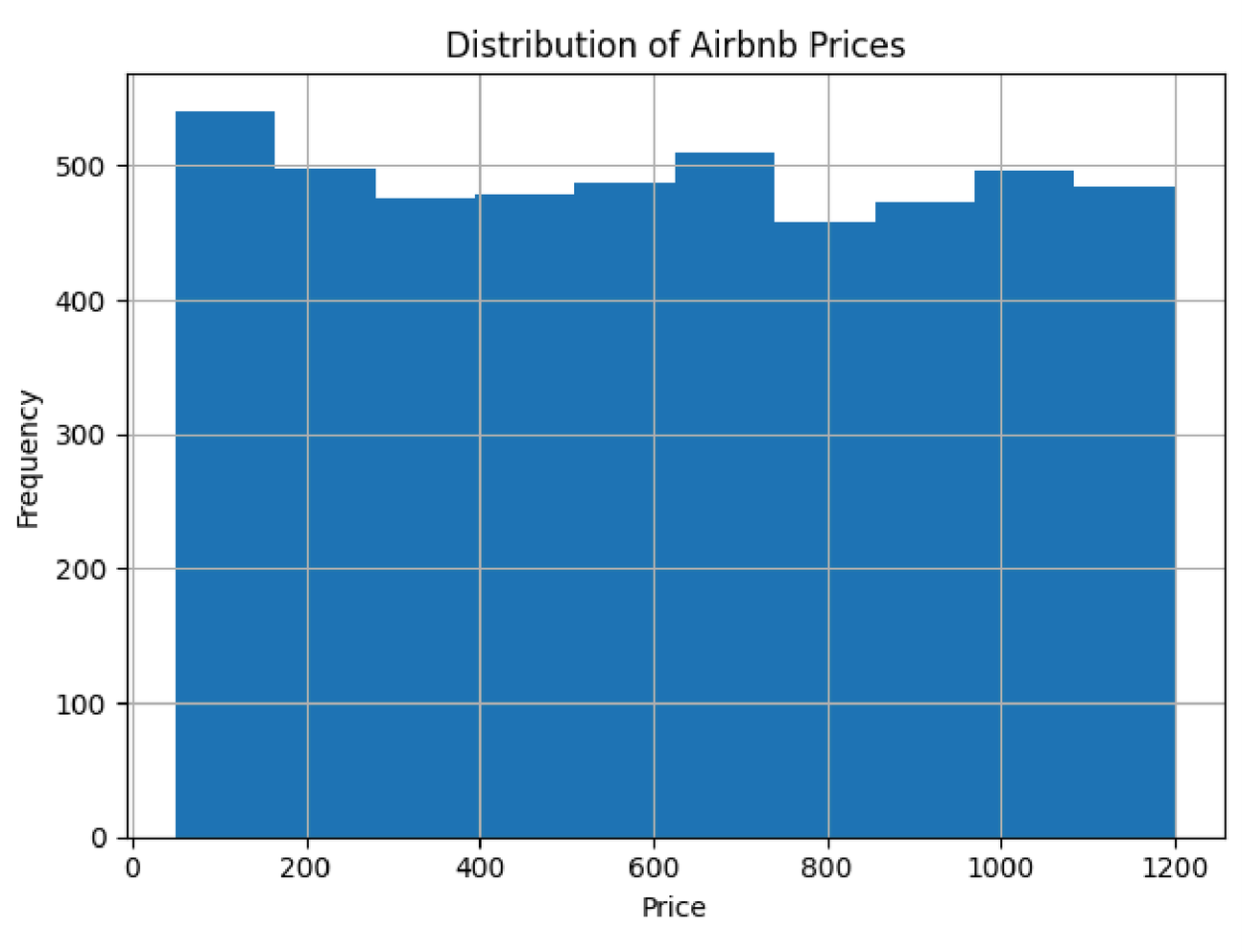
https://github.com/nma50075-star/DSCI510_FinalProject_NaMa.git

Airbnb Pricing Analysis: Key Influencing Factors

DSCI510 FINAL PROJECT

Dataset	Source	Content
Airbnb (Kaggle)	Kaggle	price, room_type
Inside Airbnb	InsideAirbnb	reviews
Tourism	US ITA	visitor arrivals

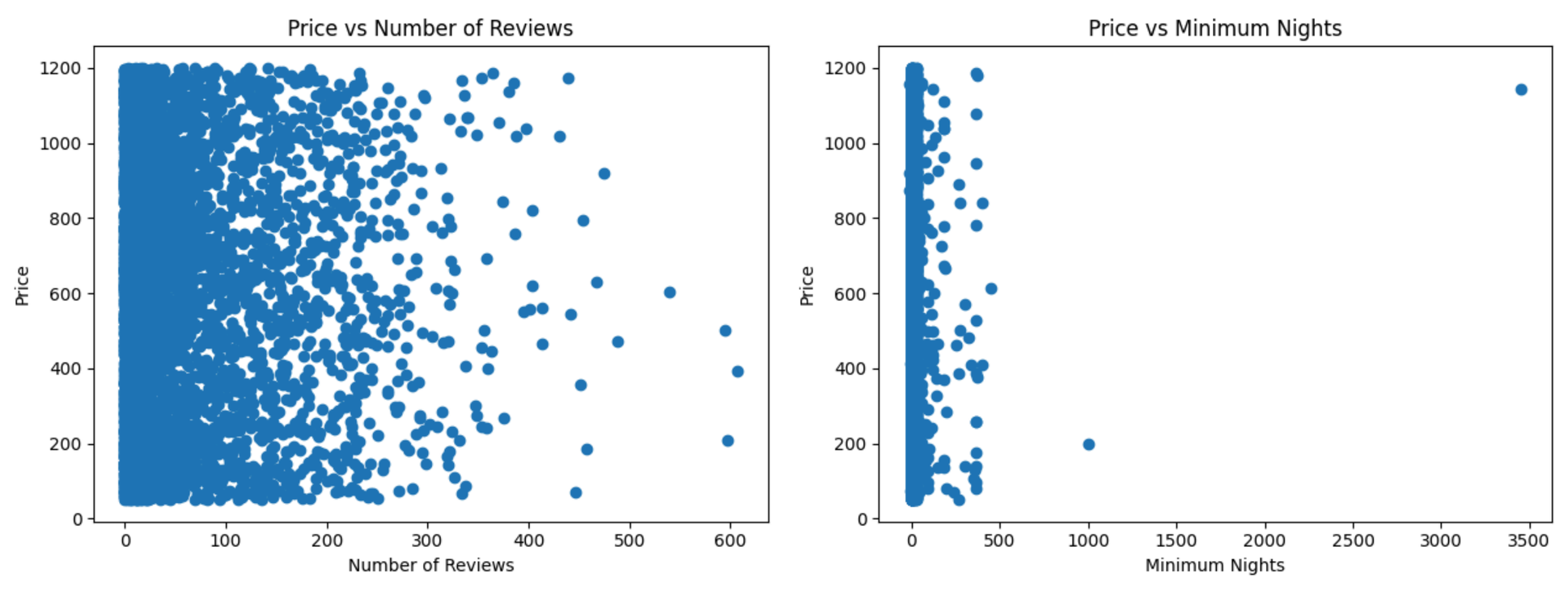
- Multiple datasets are used to capture both listing-level and market-level information
- Airbnb datasets provide pricing and listing characteristics
- Tourism data reflects broader demand trends in the travel market
- Combining these datasets allows analysis from different perspectives



Distribution of Airbnb Prices

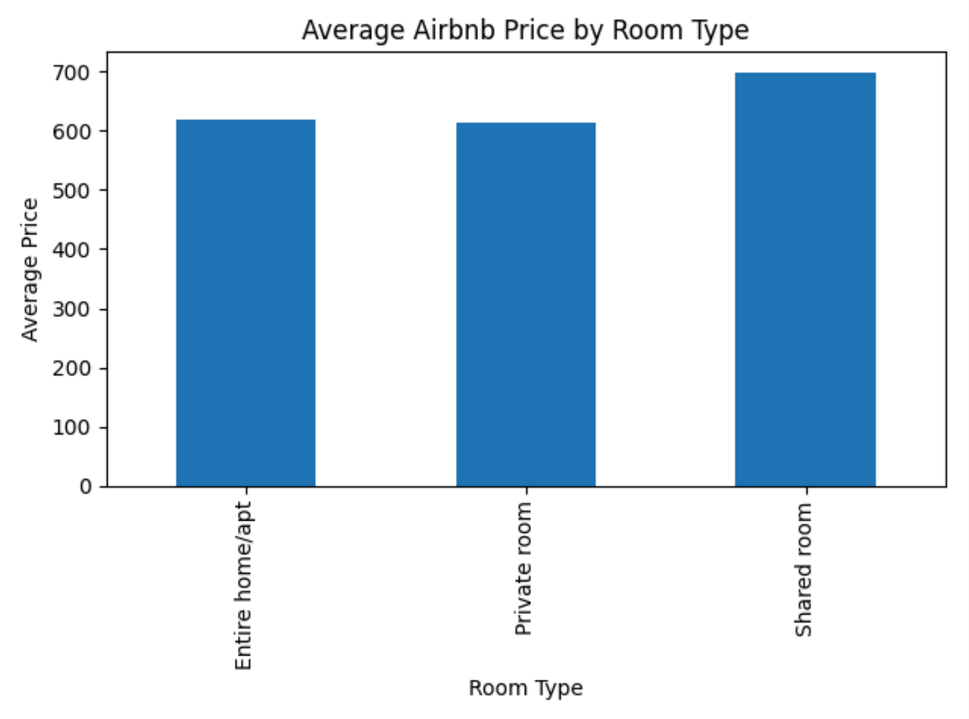
- Most listings are concentrated within a moderate price range
- Distribution is slightly right-skewed with some higher-price outliers
- Indicates a relatively consistent pricing range across listings
- without extreme price dispersion

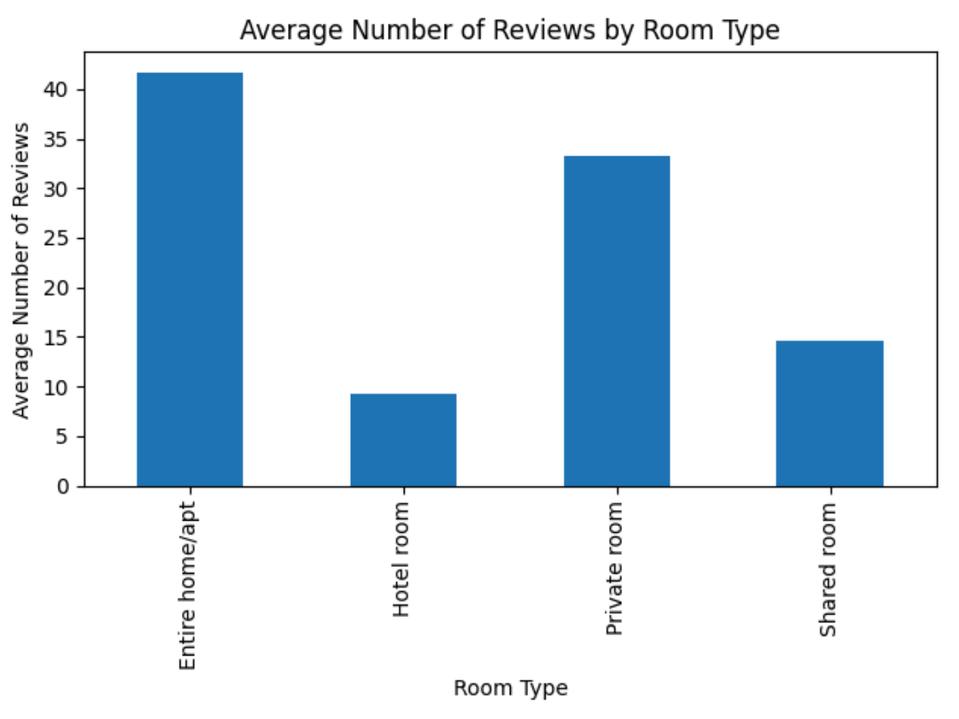
Price vs Listing Characteristics



- No strong correlation between price and number of reviews
- Listings with higher popularity do not necessarily have higher prices
- Minimum nights requirement shows limited influence on pricing

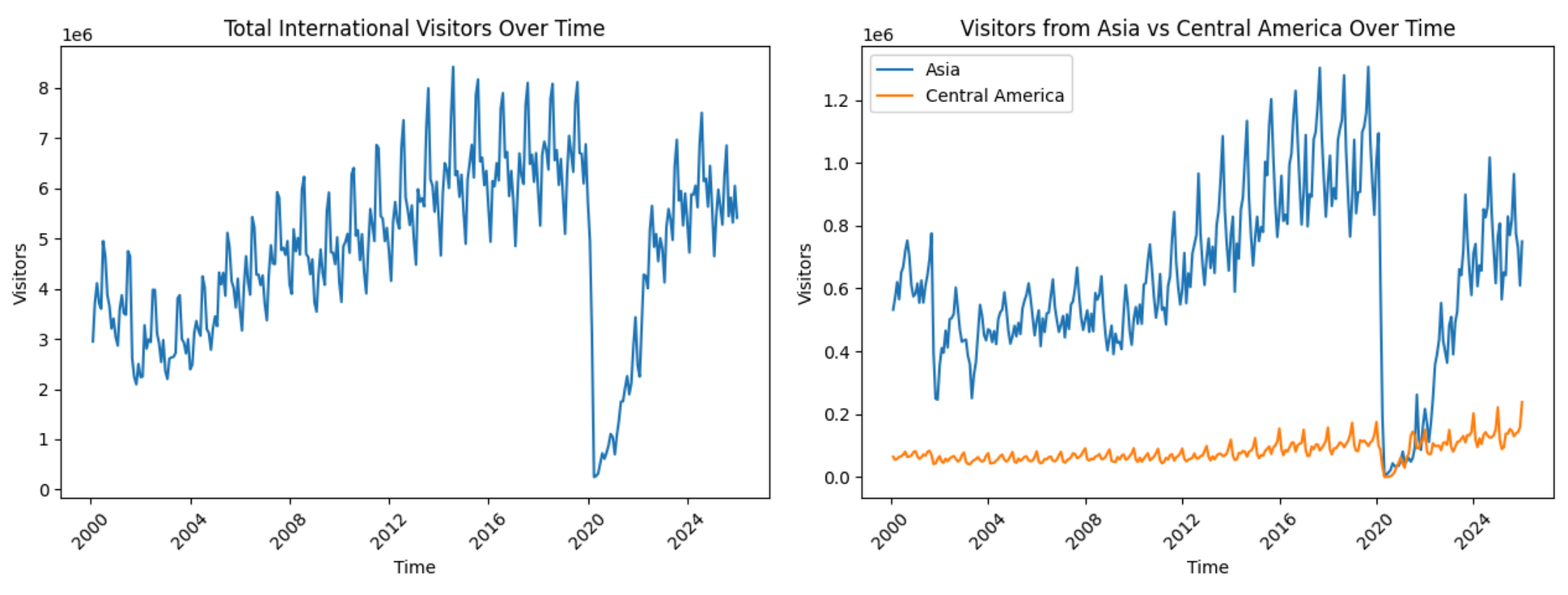
Room Type Differences





- Room type significantly affects both pricing and user engagement
- Entire homes receive the highest number of reviews
- Shared rooms show unexpected pricing patterns, possibly due to limited data or outliers.

Tourism Trends Over Time



- Tourism shows long-term growth with a sharp decline during COVID-19
- Visitor numbers begin to recover after the pandemic
- Asia contributes a larger share of visitors compared to Central America

Challenges

- Data cleaning across multiple datasets with inconsistent formats
- Handling different column naming conventions (e.g., "room type" vs "room_type")
- Dealing with missing values and outliers in pricing data
- Encoding issues when loading CSV files
- Difficulty aligning tourism data with Airbnb listing-level data
- Integrating datasets with different levels of granularity (listing vs national level)